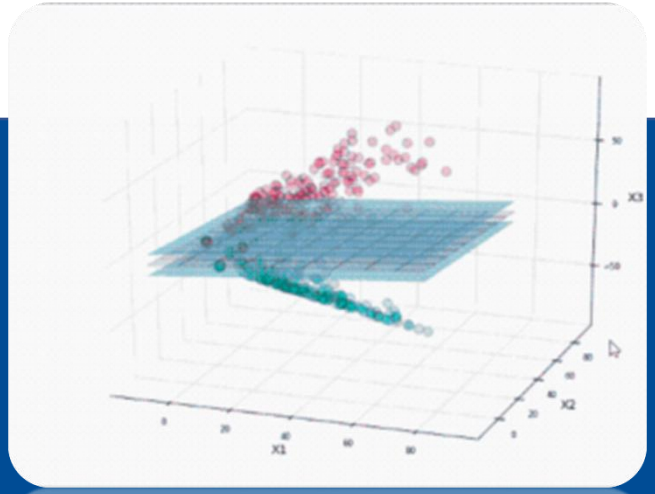
首都师范大学
CAPITAL NORMAL UNIVERSITY



第七课
支持向量机

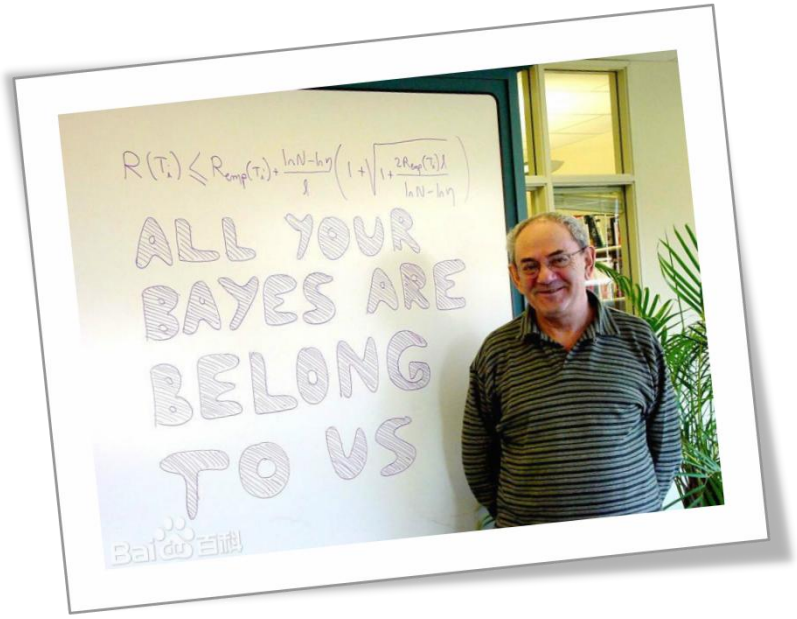
授课教师：潘贤丽

Support Vector Machine

支持向量　　机



SVM历史



推理时期	1960s	赋予机器逻辑推理能力使机器获得智能；当时的AI程序证明力一些著名的数学定理，但由于缺乏知识，远不能实现真正的智能。
知识时期	1970s	将人类的知识总结出来教给机器使机器获得智能；即“专家系统”，在很多领域获得大量进展，但由于人类知识量巨大，故出现“知识工程瓶颈”。
机器学习时期	1980s	连接主义较为流行；代表方法为神经网络。
	1990s	统计学习占据舞台。代表方法包括支持向量机等。
	21Cent.	深度神经网络被提出，连接主义卷土重来。随着数据量和计算能力的不断提升，以深度学习为基础的诸多AI应用逐渐成熟。

Support-vector networks

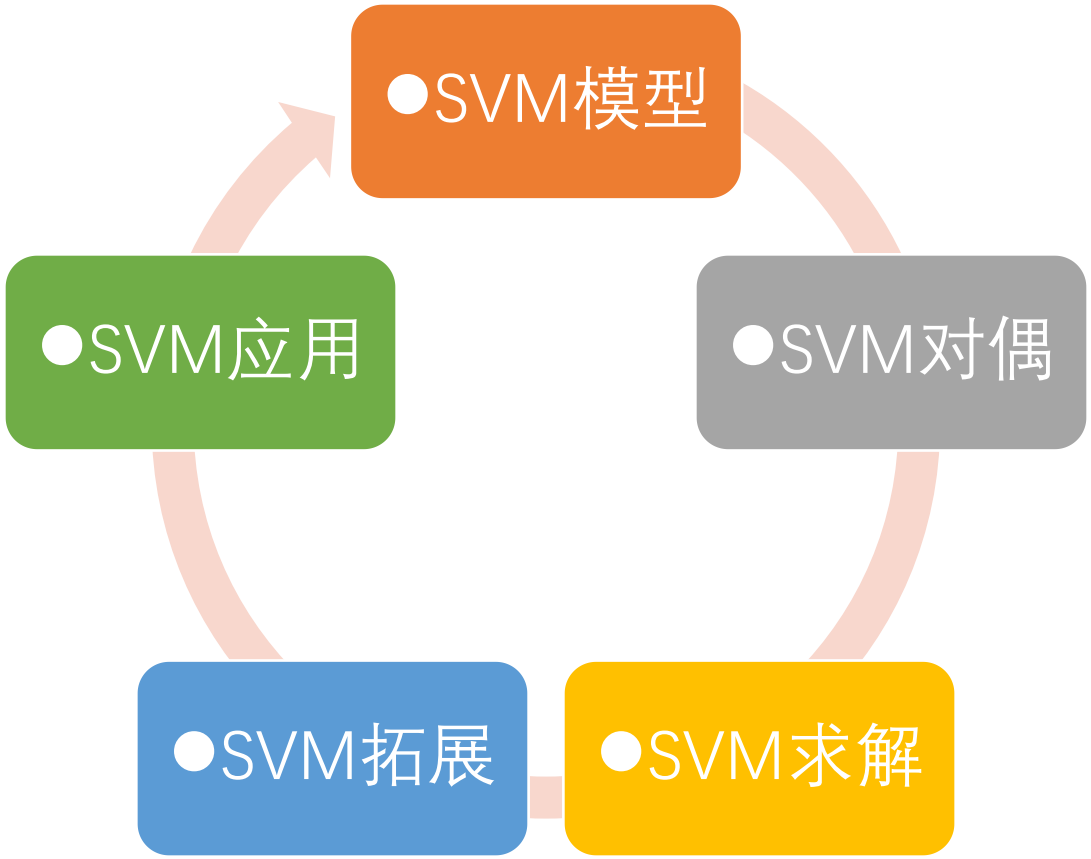
C Cortes, V Vapnik - Machine learning, 1995 - Springer

… the support-vector network was … of support-vector networks utilizing polynomial input transformations is demonstrated. We also compare the performance of the support-vector network to …

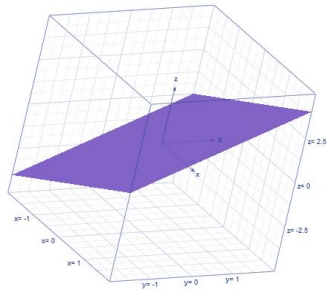
☆ 保存 99 引用 被引用次数：65944 相关文章 所有 29 个版本

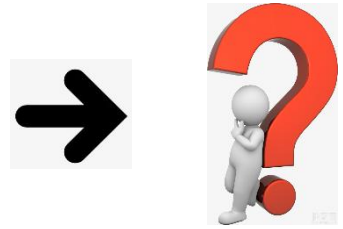
一种用于处理二分类问题的经典机器学习方法

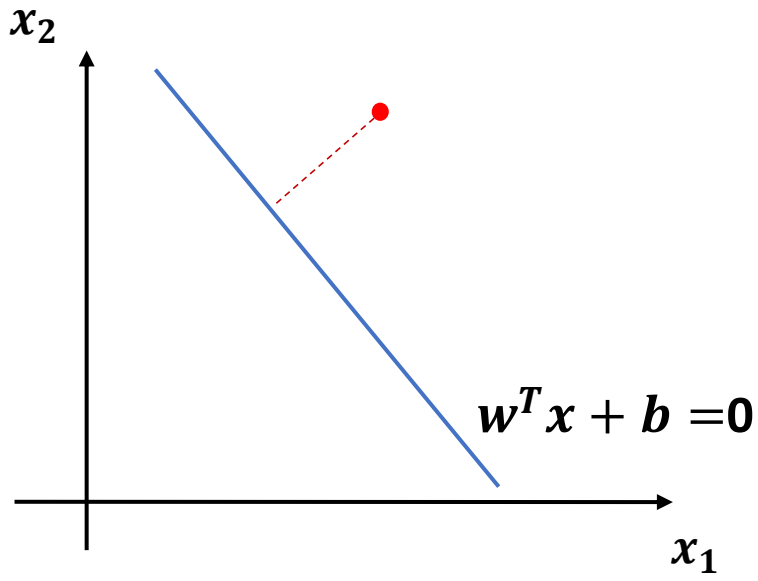
大纲



随着维度增长： 直线 → 平面 → 超平面







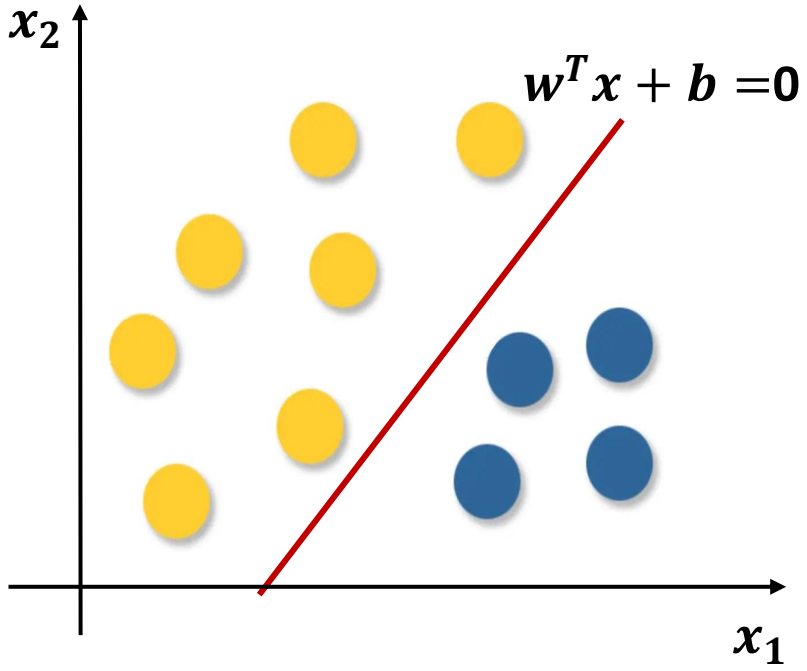
- n维空间中任意超平面的表达式：

$$w^T x + b = 0$$

- n维空间中任意点x_i到超平面的距离：

$$\frac{|w^T x_i + b|}{||w||}$$

给定训练集$T = \{(x_1, y_1), (x_2, y_2), ..., (x_l, y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1, +1\}$

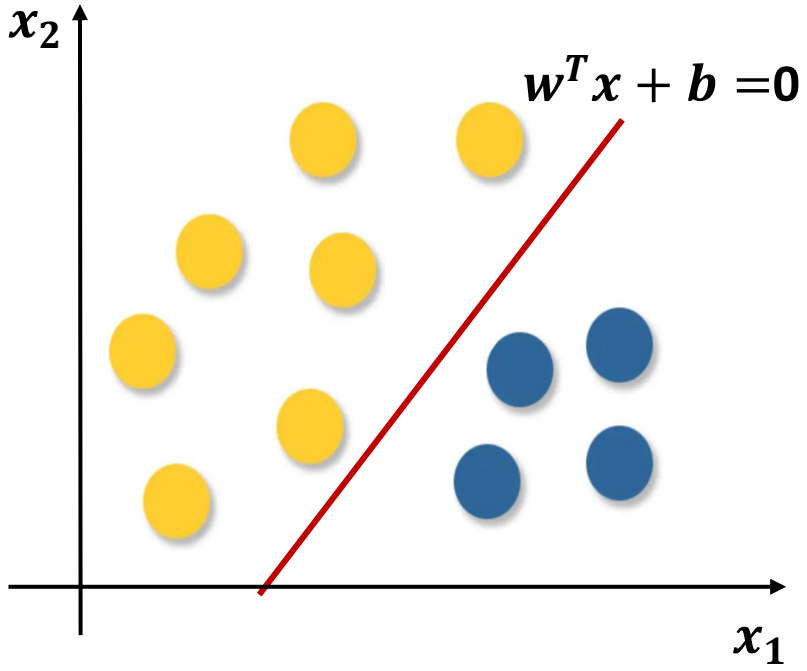


目标：

- 超平面：$w^T x + b = 0$
- 新的样本点x：

$$y = f(x) = \text{sgn}(w^T x + b)$$

给定训练集$T = \{(x_1, y_1), (x_2, y_2), ..., (x_l, y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1, +1\}$

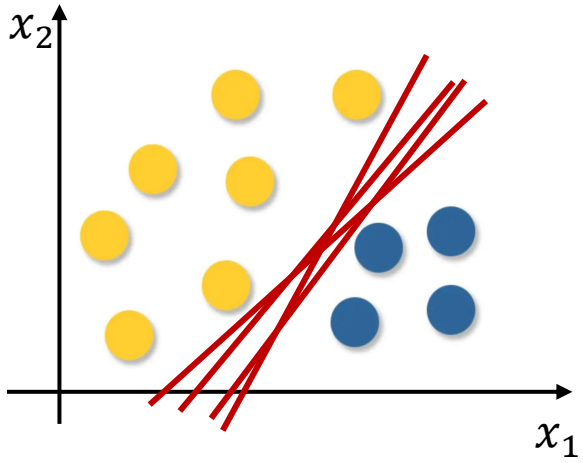


- 什么样的直线是好的？

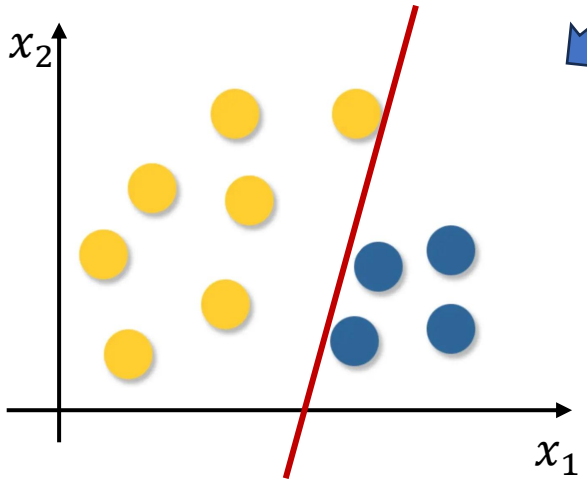
（概念性问题）

- 怎么找到这条直线？

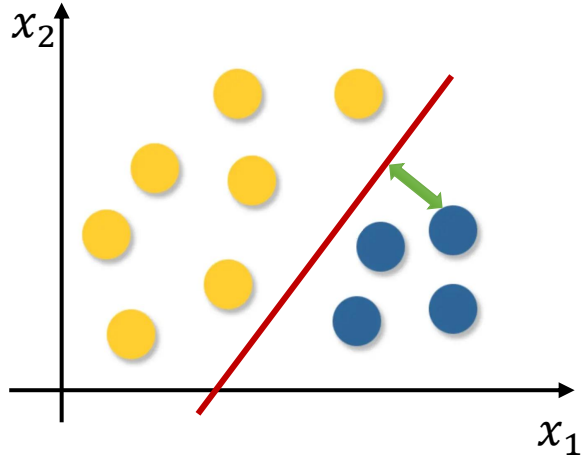
（技术性问题）



有无数条直线符合条件，哪条最好?

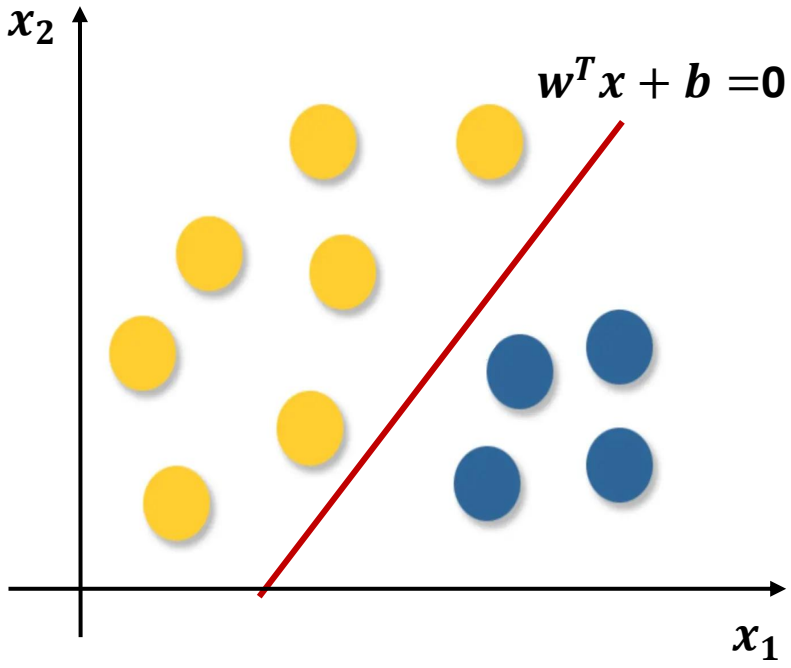


离样本近了似乎不太好~



最大间隔

给定训练集$T = \{(x_1, y_1), (x_2, y_2), ..., (x_l, y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1, +1\}$



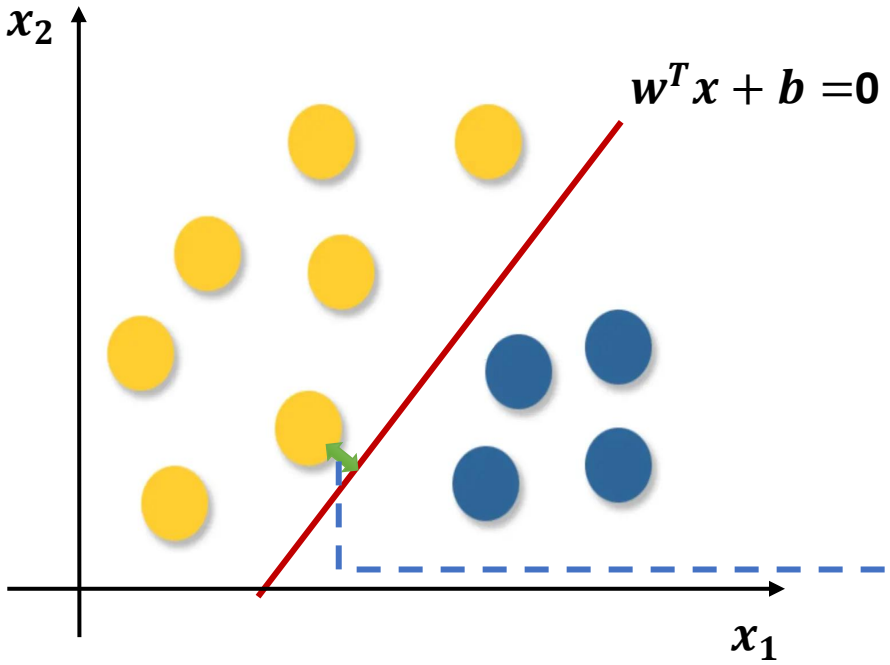
- 什么样的直线是好的？

  能实现最大间隔的直线

- 怎么找到这条直线？

  （技术性问题）

给定训练集$T=\{(x_1,y_1),(x_2,y_2),...,(x_l,y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1,+1\}$



条件1：样本要分对 $\forall i = 1,\cdots,l$

- $y_i = 1$：$(w \cdot x_i) + b \geq 0$
- $y_i = -1$：$(w \cdot x_i) + b \leq 0$

$$y_i\big((w \cdot x_i) + b\big) \geq 0$$

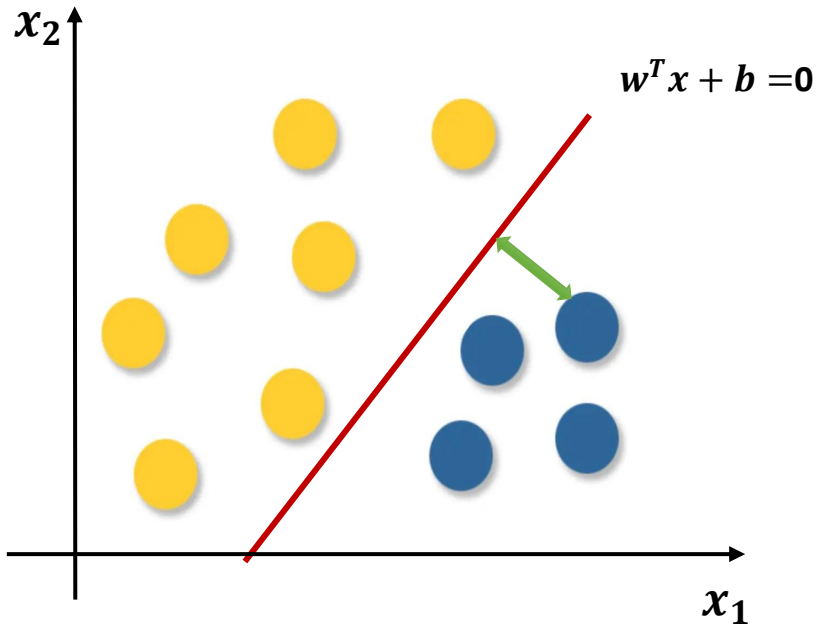
条件2：最大间隔

$$\gamma = \min_i \gamma_i = \frac{|w^T x_i + b|}{||w||}$$

$$\max_{w,b} \gamma$$

$$\max_{w,b} \gamma \quad s.t. \quad \frac{y_i\big((w \cdot x_i) + b\big)}{||w||} \geq \gamma$$

给定训练集$T = \{(x_1, y_1), (x_2, y_2), ..., (x_l, y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1, +1\}$



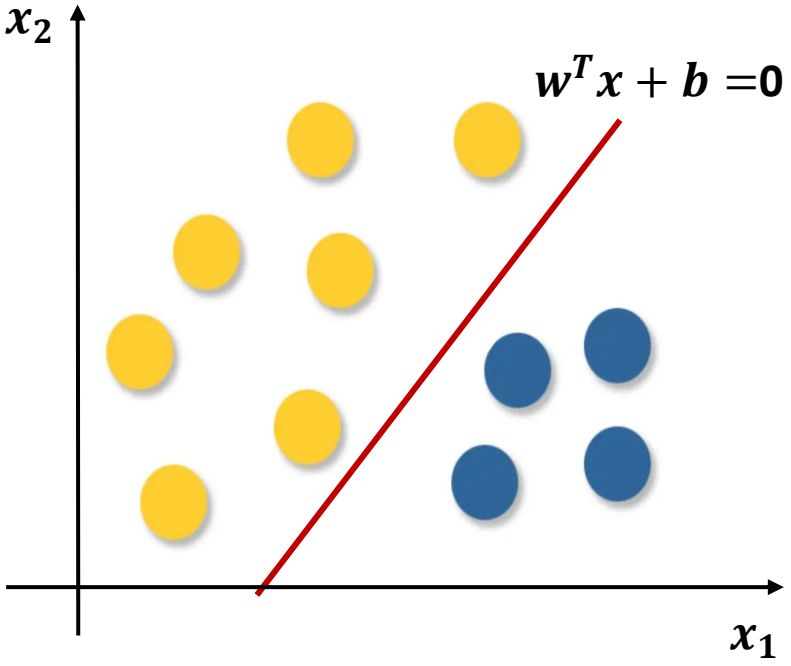
$$\max_{w,b} \gamma$$
$$s.t. \ \frac{y_i((w \cdot x_i) + b)}{||w||} \geq \gamma$$

$$\max_{w,b} \frac{\tilde{\gamma}}{||w||}$$
$$s.t. \ y_i((w \cdot x_i) + b) \geq \tilde{\gamma}$$

$\tilde{\gamma} = 1$

$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i((w \cdot x_i) + b) \geq 1, \ i = 1, \cdots, l$$

给定训练集$T = \{(x_1, y_1), (x_2, y_2), \ldots, (x_l, y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1, +1\}$



- 什么样的直线是好的？

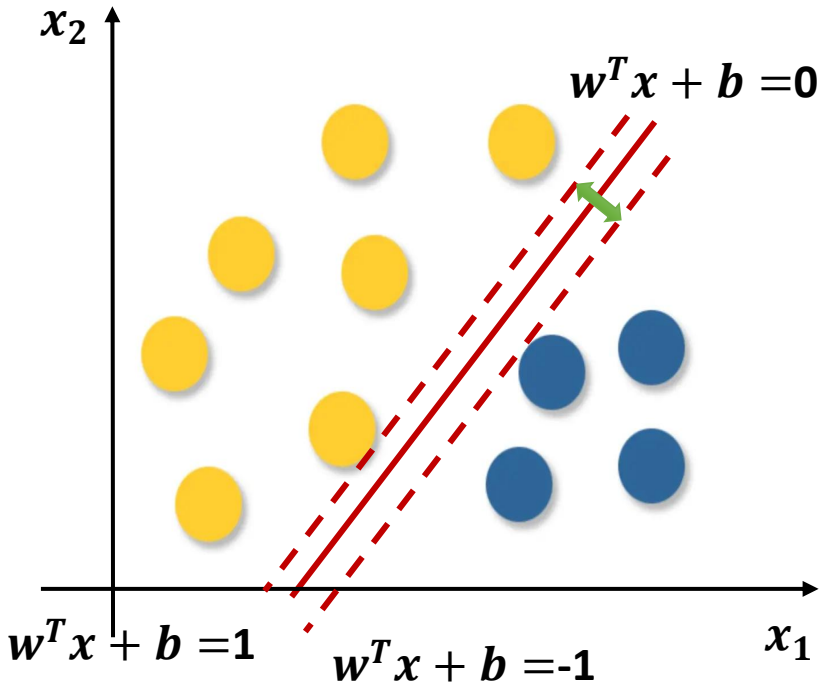
能实现最大间隔的直线

- 怎么找到这条直线？

在样本全都分对的前提下，最大化所有样本到直线的最小距离

$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i\big((w \cdot x_i) + b\big) \geq 1, \ i = 1, \cdots, l$$

给定训练集$T = \{(x_1, y_1), (x_2, y_2), \ldots, (x_l, y_l)\}$，其中$x_i \in R^n$，$y_i \in \{-1, +1\}$

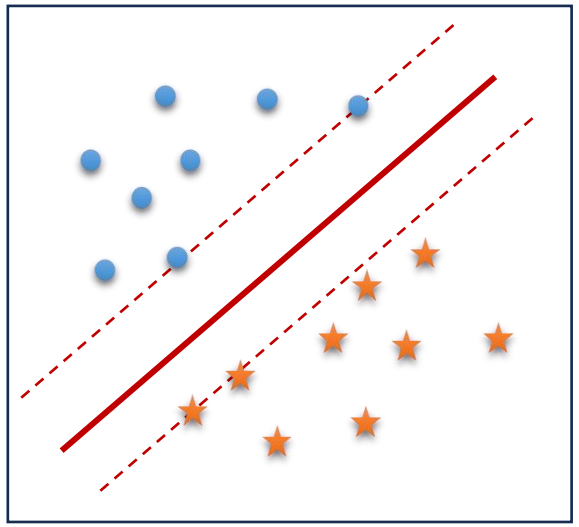


$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i\big((w \cdot x_i) + b\big) \geq 1, \ i = 1, \cdots, l$$

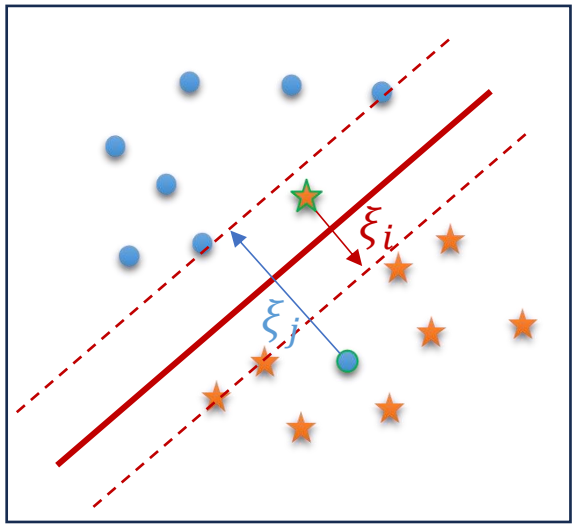
- 决策超平面：$w^T x + b = 0$
- 支持超平面：$w^T x + b = \pm 1$
- 间隔：$\frac{2}{||w||}$

SVM模型

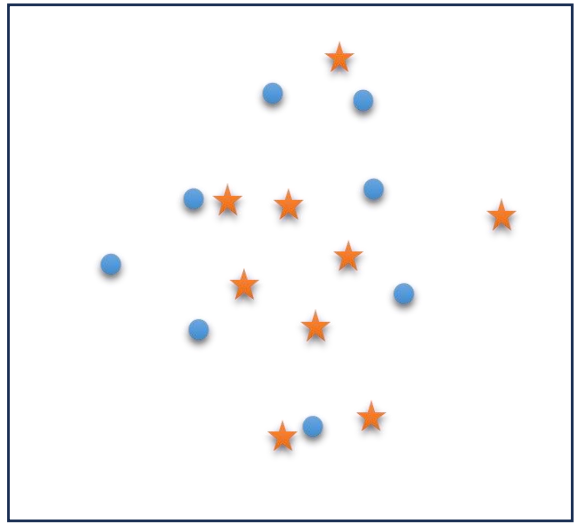
线性可分



近似线性可分
（线性不可分）



非线性可分



硬间隔

$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i\big((w \cdot x_i) + b\big) \geq 1, \ i = 1, \cdots, l$$

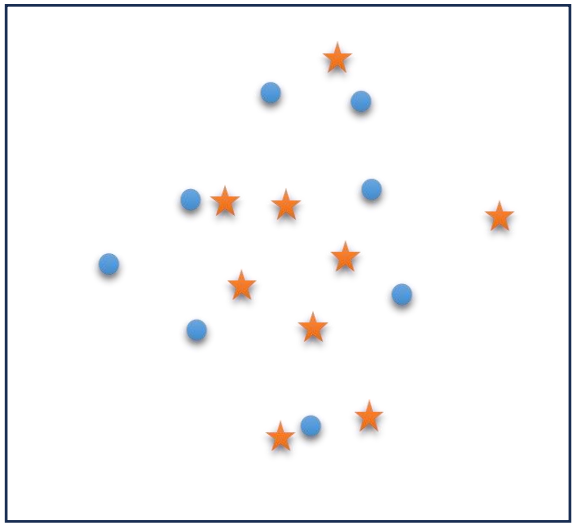
软间隔

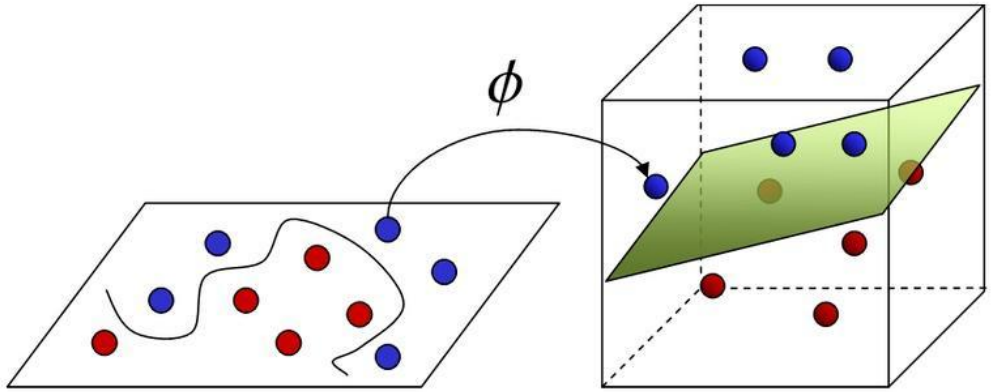
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$

松弛变量

$$s.t. \ y_i(w^T x_i + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

非线性可分





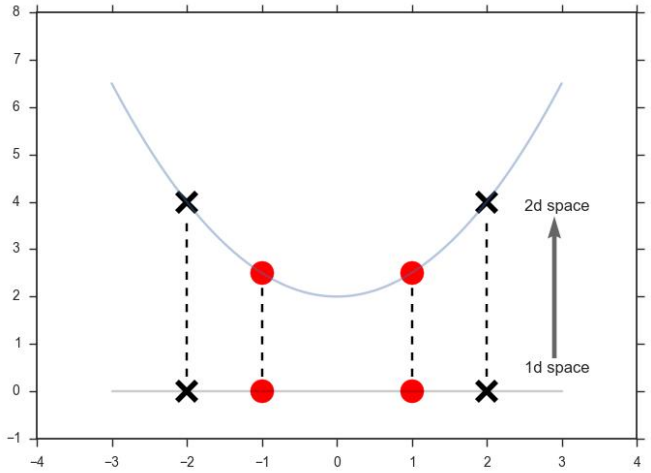
把样本映射到高维空间，让它近似线性可分：$x \to \Phi(x)$

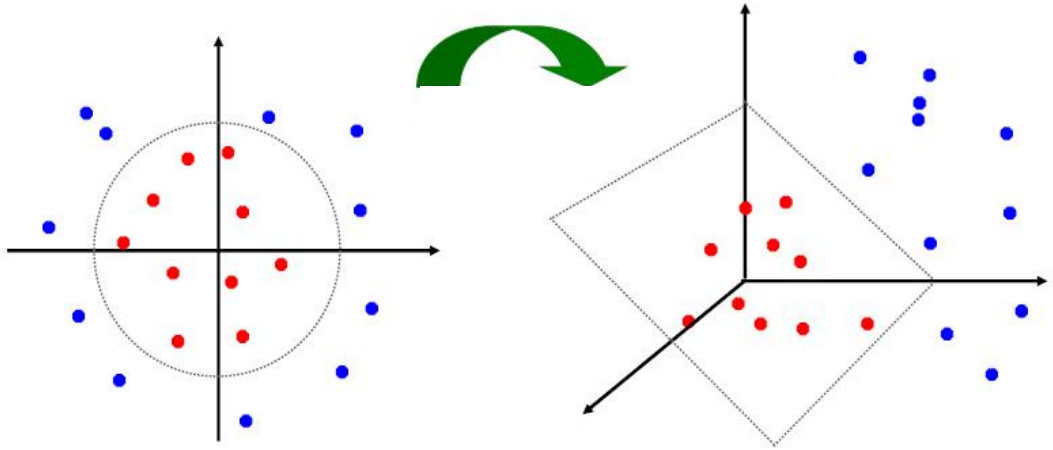
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$

$$s.t.\ y_i(w^T\Phi(x_i) + b) \geq 1 - \xi_i, i = 1, \ldots, l$$

$$\xi_i \geq 0,\ i = 1, \ldots, l$$

SVM模型



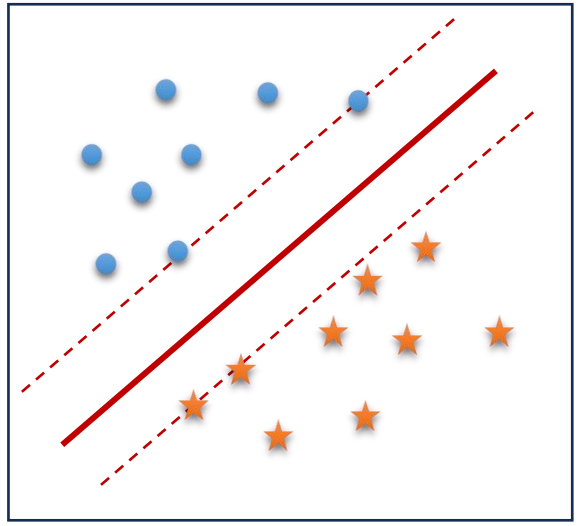




映射Φ怎么选？

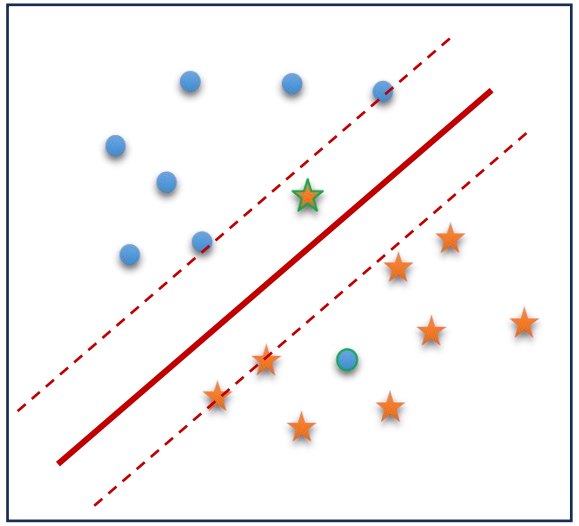
SVM模型

线性可分



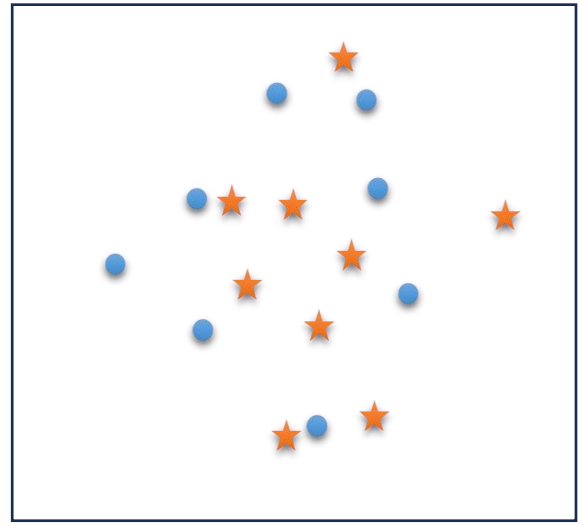
$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i\big((w \cdot x_i) + b\big) \geq 1, \quad i = 1, \cdots, l$$

近似线性可分（线性不可分）



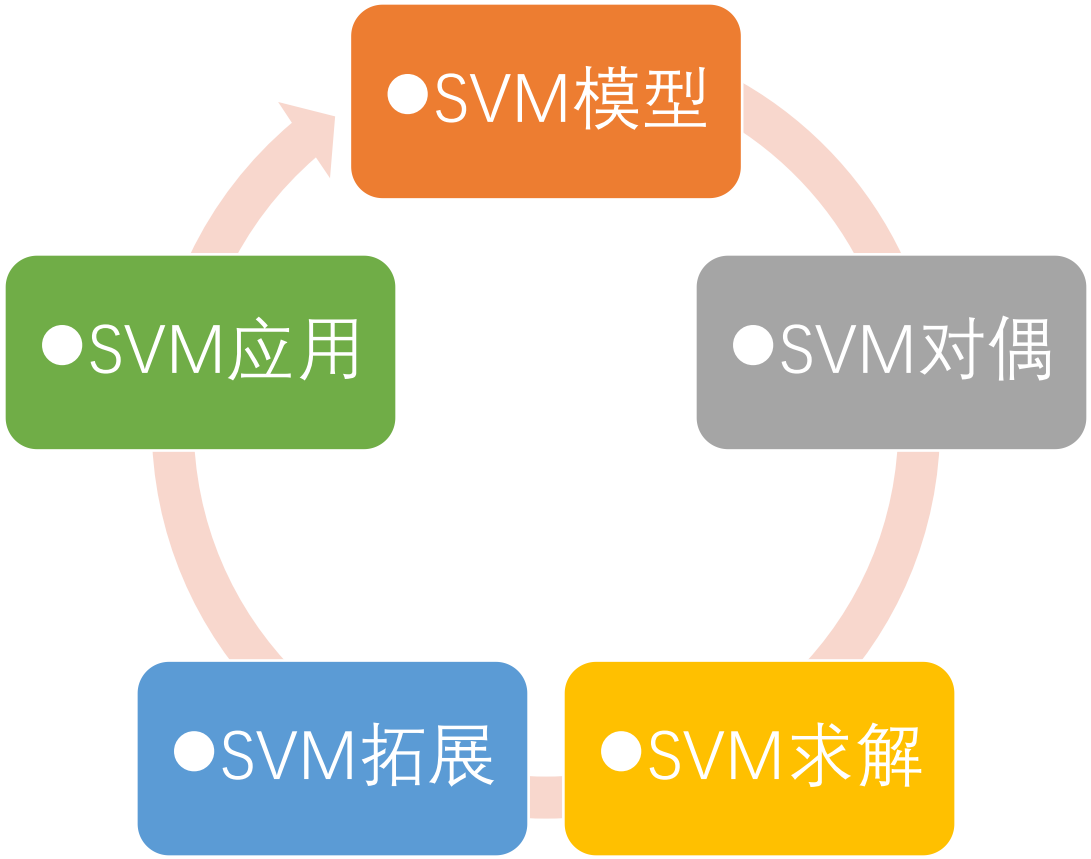
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\, y_i(w^T x_i + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

非线性可分



$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\, y_i(w^T \Phi(x_i) + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

大纲



原问题

$$\begin{aligned} \min_{x} \quad & f_0(x) \\ s.t. \quad & f_i(x) \le 0, i = 1, 2, \dots, m \\ & h_j(x) = 0, j = 1, 2, \dots, p \end{aligned}$$

$$L(x, \lambda, v) = f_0(x) + \sum_{i=1}^{m} \lambda_i f_i(x) + \sum_{j=1}^{p} v_j h_j(x)$$

$$g(\lambda, v) = \inf_{x} L(x, \lambda, v)$$

对偶问题

$$\begin{aligned} \max_{\lambda, v} \quad & g(\lambda, u) \\ s.t. \quad & \lambda \ge 0 \end{aligned}$$

原问题：

- 约束太复杂，不好求解
- 非凸/不知道凸不凸

对偶问题：

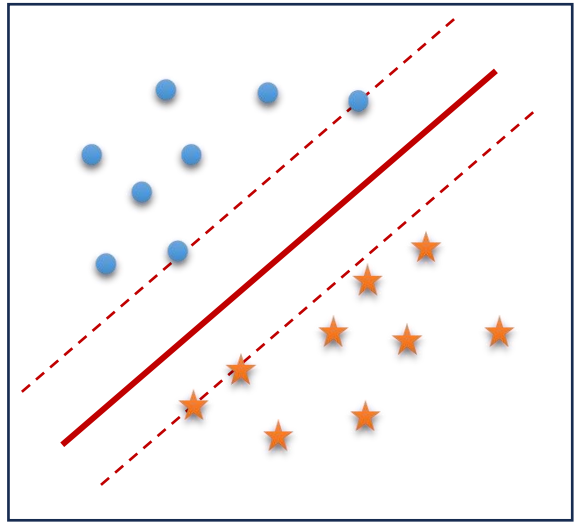
- 约束简单，好求解
- 凸优化

最优性条件（KKT条件）

$$\begin{cases} \dfrac{\partial L(x, \lambda^*, v^*)}{\partial x}\Big|_{x=x^*} = 0 \\ \lambda_i^* f_i(x^*) = 0 \\ f_i(x^*) \le 0, \ h_j(x^*) = 0 \\ \lambda_i^* \ge 0 \end{cases}$$

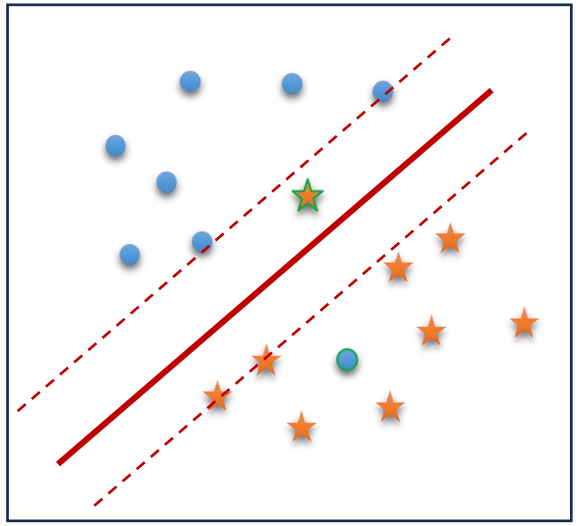
SVM模型

线性可分



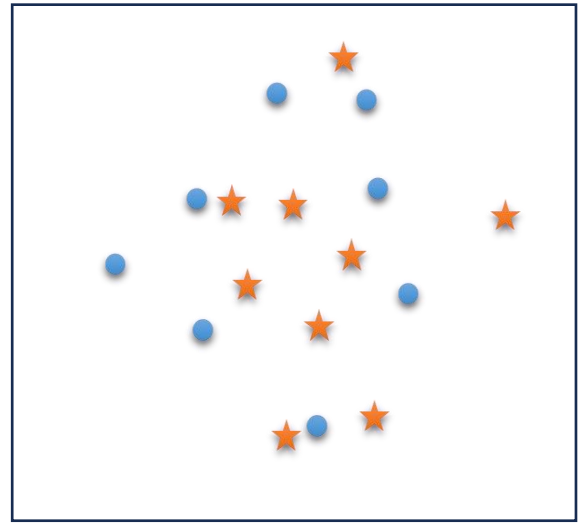
$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i\big((w \cdot x_i) + b\big) \geq 1, \quad i = 1, \cdots, l$$

近似线性可分（线性不可分）



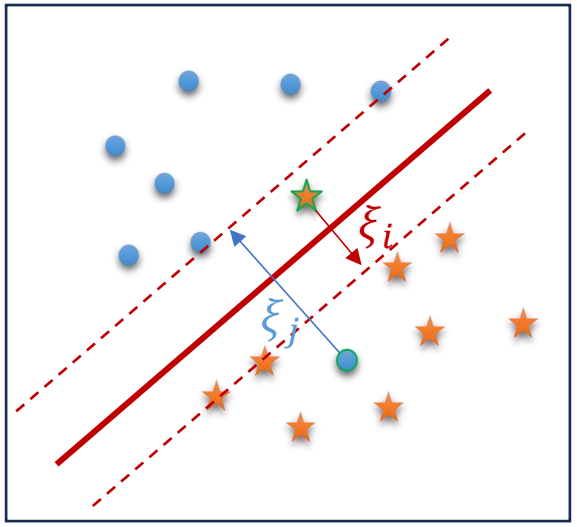
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\, y_i(w^T x_i + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

非线性可分



$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\, y_i(w^T \Phi(x_i) + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

近似线性可分（线性不可分）



$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T x_i + b) \geq 1 - \xi_i,\ i = 1, \ldots, l$$
$$\xi_i \geq 0,\ i = 1, \ldots, l$$

引入拉格朗日函数：

$$L = \frac{1}{2}||w||^2 + C\sum_{i=1}^{l}\xi_i - \sum_{i=1}^{l}\alpha_i(y_i(w \cdot x_i + b) - 1 + \xi_i) - \sum_{i=1}^{l}\beta_i\xi_i$$

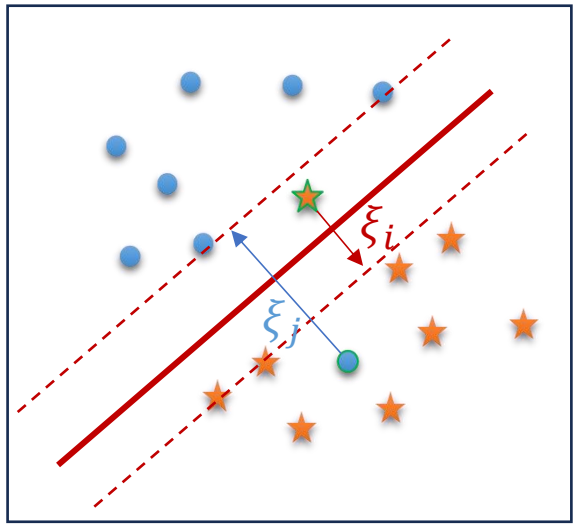
$L(w, b, \alpha)$分别关于$w,\ b, \xi_i$求导：

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{l}\alpha_i y_i x_i = 0 \Rightarrow w = \sum_{i=1}^{l}\alpha_i y_i x_i$$
$$\frac{\partial L}{\partial b} = \sum_{i=1}^{l}\alpha_i y_i = 0$$
$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \beta_i = 0 \Rightarrow 0 \leq \alpha_i \leq C$$

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j (x_i \cdot x_j)\alpha_i\alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t. \ \sum_{i=1}^{l} y_i\alpha_i = 0$$
$$0 \leq \alpha_i \leq C,\ i = 1, \ldots, l.$$

SVM对偶

近似线性可分（线性不可分）



原问题

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T x_i + b) \geq 1 - \xi_i,\ i = 1, \ldots, l$$
$$\xi_i \geq 0,\ i = 1, \ldots, l$$

b怎么求？

$$\boldsymbol{w = \sum_{i=1}^{l} \alpha_i y_i x_i}$$

对偶问题

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j (x_i \cdot x_j)\alpha_i \alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t. \sum_{i=1}^{l} y_i \alpha_i = 0$$
$$0 \leq \alpha_i \leq C,\ i = 1, \ldots, l.$$

SVM对偶

原问题

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T x_i + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0,\ i = 1, \ldots, l$$

对偶问题

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j (x_i \cdot x_j)\alpha_i\alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t.\ \sum_{i=1}^{l} y_i\alpha_i = 0$$
$$0 \leq \alpha_i \leq C,\ i = 1, \ldots, l.$$

KKT条件

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{l}\alpha_i y_i x_i = 0 \rightarrow w = \sum_{i=1}^{l}\alpha_i y_i x_i$$

$$\frac{\partial L}{\partial b} = \sum_{i=1}^{l}\alpha_i y_i = 0$$

$$\frac{\partial L}{\partial \xi_i} = C - \alpha_i - \beta_i = 0 \rightarrow 0 \leq \alpha_i \leq C$$

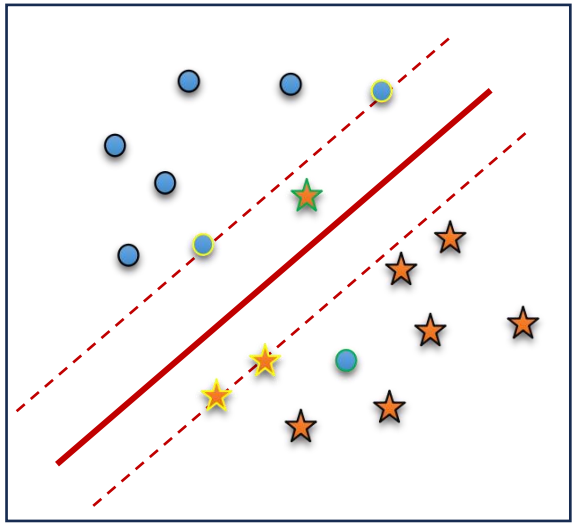
$$\alpha_i(y_i(w \cdot x_i + b) - 1 + \xi_i) = 0$$
$$\beta_i\xi_i = 0$$

互补松弛条件

$$\alpha_i \geq 0, \beta_i \geq 0$$
$$y_i(w \cdot x_i + b) \geq 1 - \xi_i$$
$$\xi_i \geq 0$$

找$0 < \alpha_i < C$对应的样本：$b = y_i - w \cdot x_i,\ \forall i: 0 < \alpha_i < C$

近似线性可分（线性不可分）



$$w = \sum_{i=1}^{l} \alpha_i y_i x_i$$

$0 \le \alpha_i \le C$

- $\alpha_i = 0$：落在支持超平面上以及外侧的样本
- $0 < \alpha_i < C$：落在支持超平面上的样本
- $\alpha_i = C$：落在支持超平面上以及内侧的样本

支持向量：$\alpha_i \neq 0$对应的样本

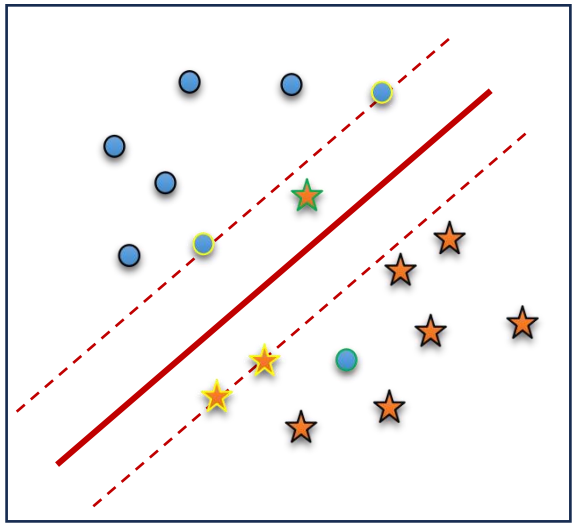
SVM对偶

原问题

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T x_i + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0,\ i = 1, \ldots, l$$

对偶问题

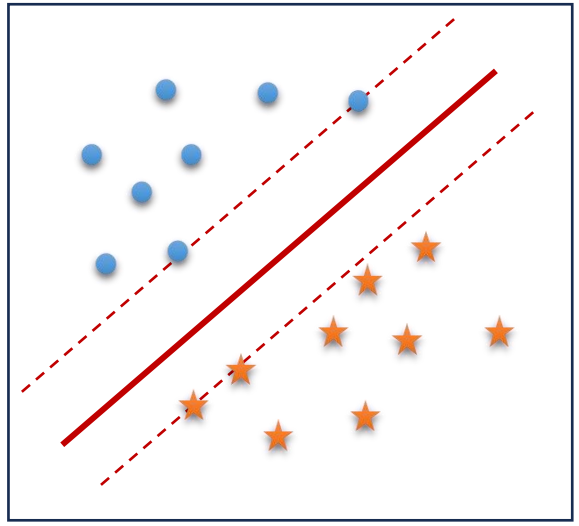
$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j (x_i \cdot x_j)\alpha_i\alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t.\ \sum_{i=1}^{l} y_i\alpha_i = 0$$
$$0 \leq \alpha_i \leq C,\ i = 1, \ldots, l.$$



- 对偶问题更容易求解
- 对偶解有很好的几何含义
- 对偶解有稀疏性（支持向量）

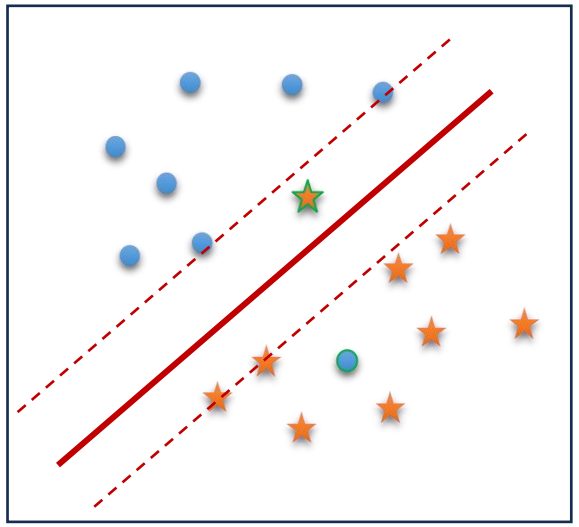
SVM对偶

线性可分



$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \ \ y_i\big((w \cdot x_i) + b\big) \geq 1,$$
$$i = 1, \cdots, l$$

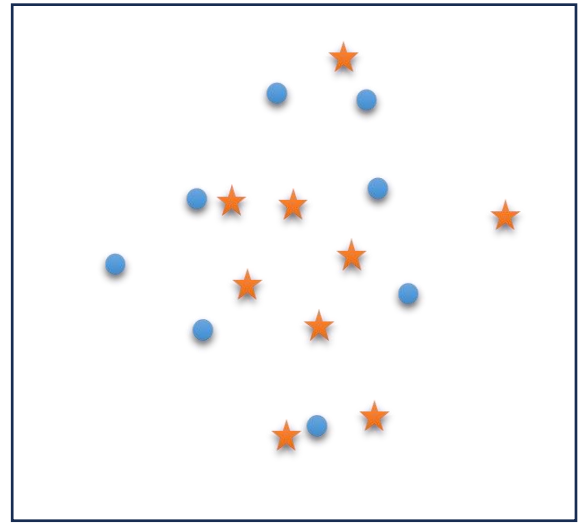
近似线性可分（线性不可分）



$$\min_{w,b,\xi} \ \ \frac{1}{2}||w||^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T x_i + b) \geq 1 - \xi_i, \ i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

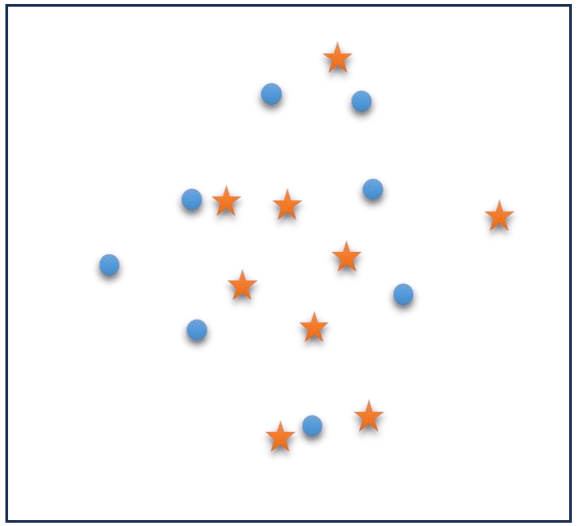
对偶

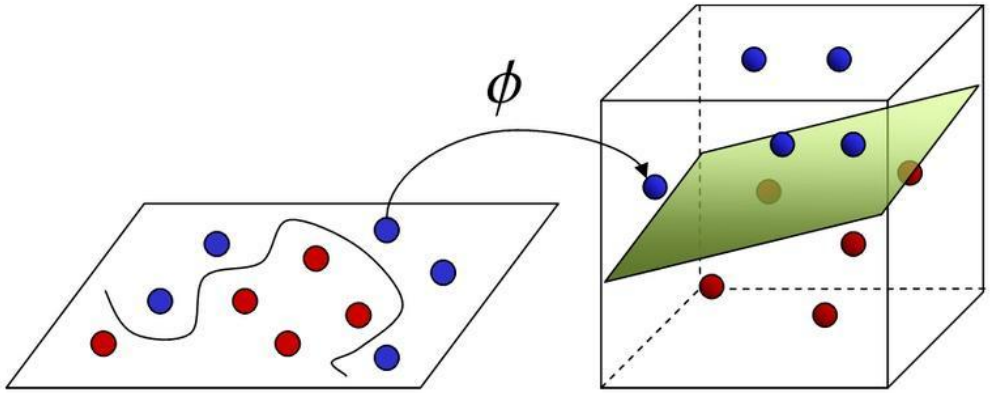
非线性可分



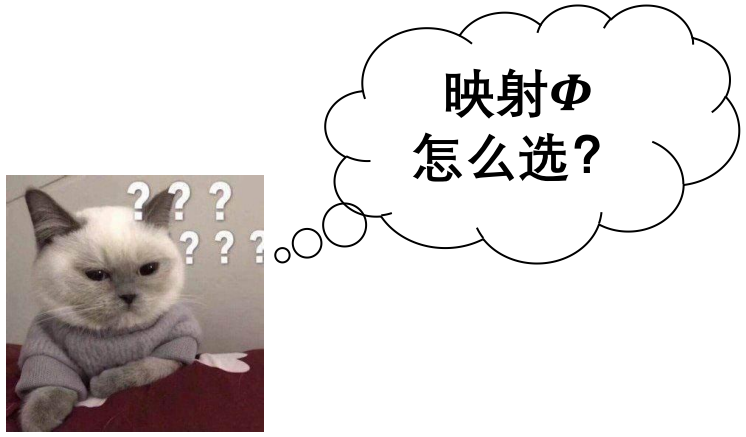
$$\min_{w,b,\xi} \ \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T \boldsymbol{\Phi}(x_i) + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

非线性可分





把样本映射到高维空间，让它近似线性可分：$x \to \Phi(x)$



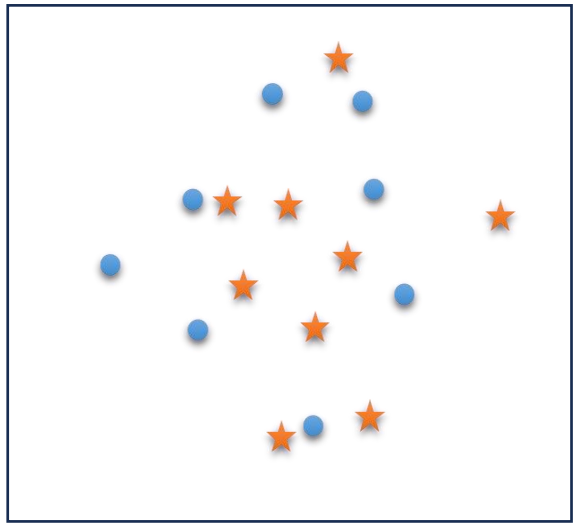
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$

$$s.t.\ y_i(w^T\Phi(x_i) + b) \geq 1 - \xi_i,\ i = 1, \ldots, l$$

$$\xi_i \geq 0,\ i = 1, \ldots, l$$

SVM对偶

非线性可分



$$f(x) = w^T \Phi(x) + b$$

$$= \sum_{i=1}^{l} \alpha_i y_i (\Phi(x_i) \cdot \Phi(x)) + b$$

原问题

$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l} \xi_i$$

$$s.t.\ y_i(w^T \Phi(x_i) + b) \geq 1 - \xi_i, i = 1, \ldots, l$$

$$\xi_i \geq 0,\ i = 1, \ldots, l$$

$$w = \sum_{i=1}^{l} \alpha_i y_i \Phi(x_i)$$

对偶问题

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j (\Phi(x_i) \cdot \Phi(x_j)) \alpha_i \alpha_j + \sum_{j=1}^{l} \alpha_j$$

$$s.t. \ \sum_{i=1}^{l} y_i \alpha_i = 0$$

$$0 \leq \alpha_i \leq C,\ i = 1, \ldots, l.$$

核函数

定义：设X是输入空间，H为特征空间，如果存在一个从X到H的映射：

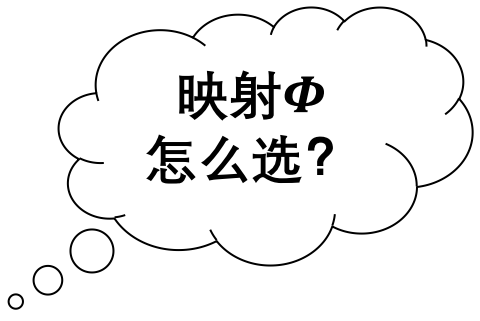
$$\Phi:\ X \longrightarrow H$$

使得对所有$x, z \in X$，函数$K(x, z)$满足条件：

$$K(x, z)=\Phi(x) \cdot \Phi(z)$$

则称$K(x, z)$为核函数，$\Phi(x)$为映射函数.









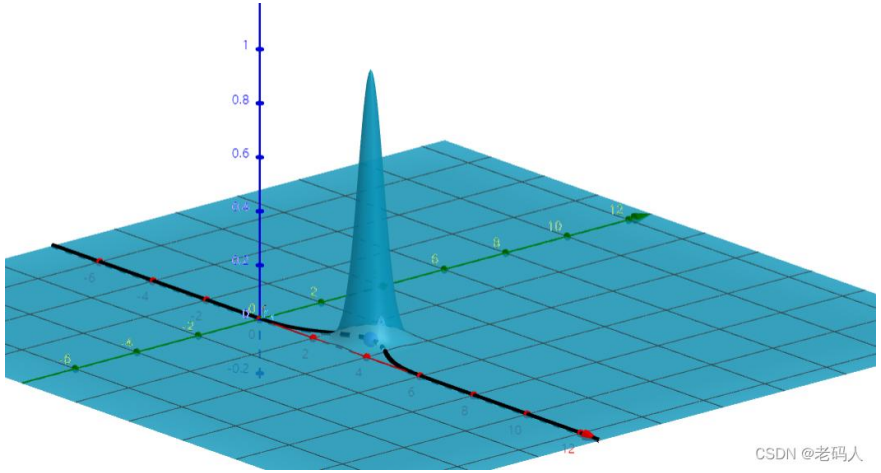
核函数

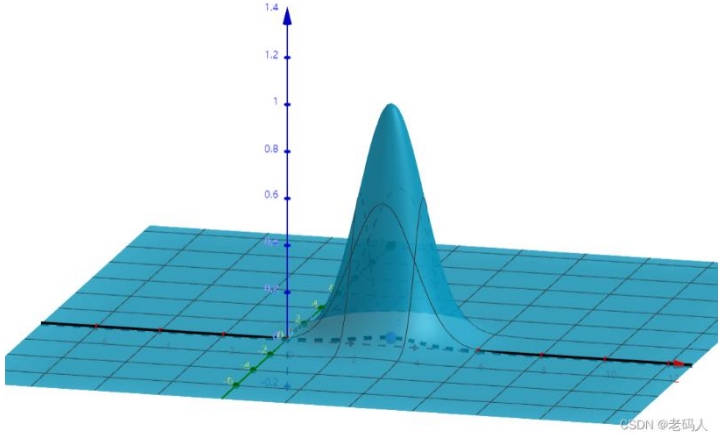
Mercer定理: 任何半正定对称函数都可以作为核函数

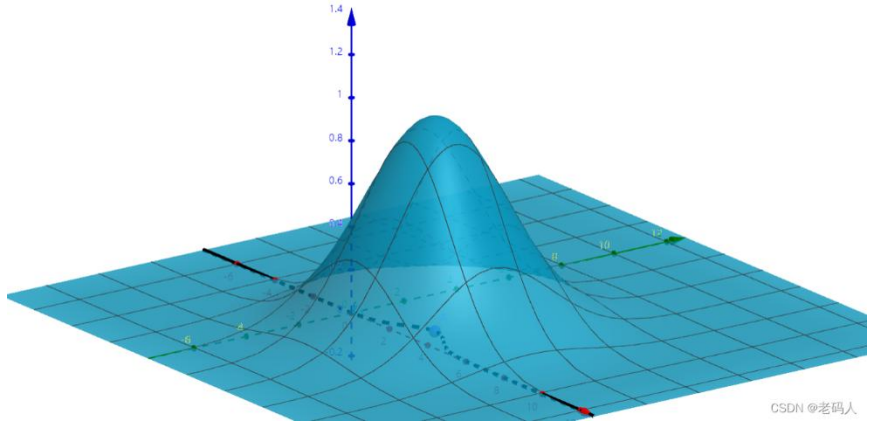
- **线性核：** $K(x_i, x_j) = x_i^T x_j$
- **多项式核：** $K(x_i, x_j) = (x_i^T x_j)^d$ （$d \geq 1$为多项式的次数）
- **高斯核：** $K(x_i, x_j) = \exp(-\frac{||x_i - x_j||^2}{2\sigma^2})$ （$\sigma > 0$为高斯核的带宽）
- 拉普拉斯核： $K(x_i, x_j) = \exp(-\frac{||x_i - x_j||}{\sigma})$ （$\sigma > 0$）
- Sigmoid核： $K(x_i, x_j) = \tanh(\beta x_i^T x_j + \theta)$（$\beta > 0$，$\theta < 0$）

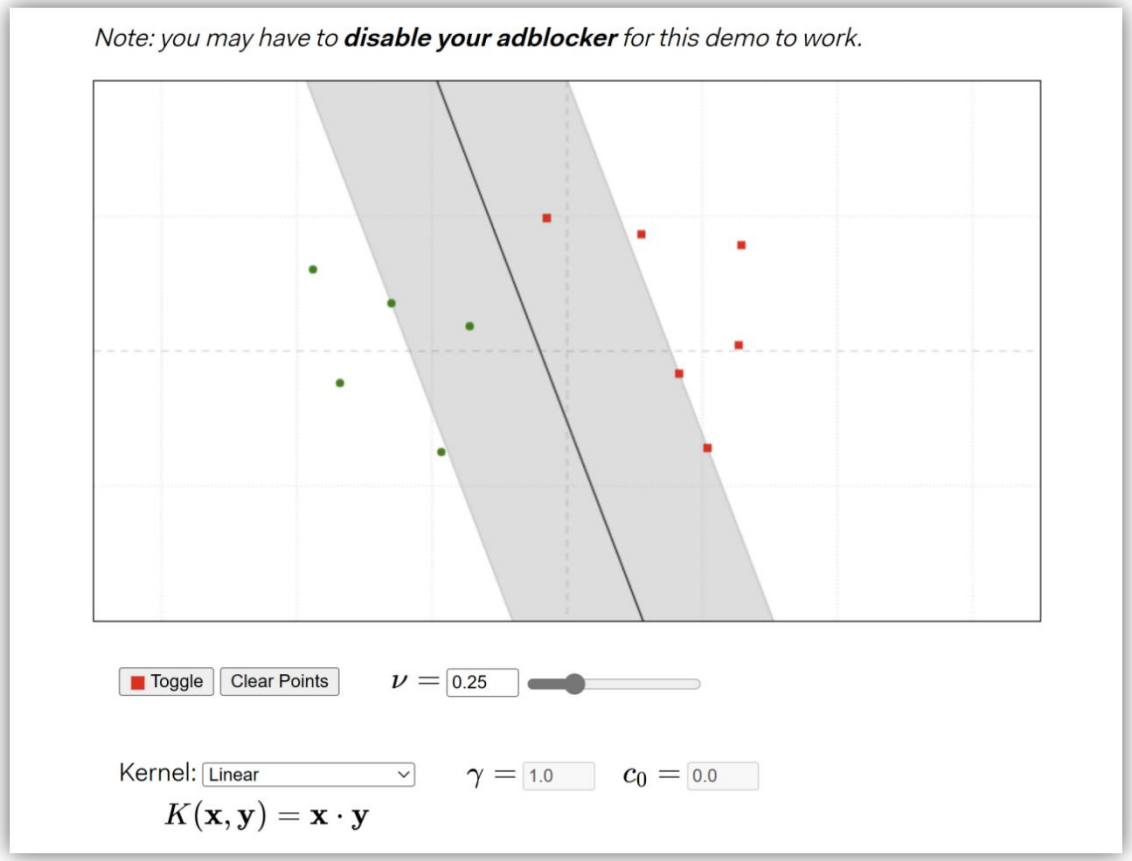
核函数

高斯核： $K(x_i, x_j) = \exp(-\frac{||x_i - x_j||^2}{2\sigma^2})$



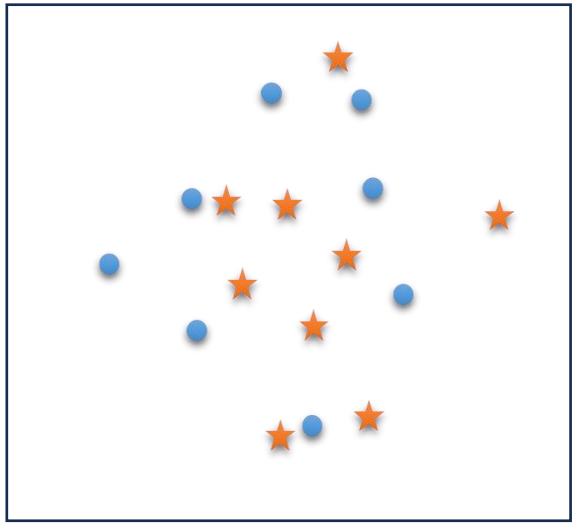






[Interactive demo of Support Vector Machines (SVM) (greitemann.dev)]

SVM对偶



对偶问题

$$\max_{\alpha} \quad -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j K(x_i, x_j)\alpha_i\alpha_j + \sum_{j=1}^{l}\alpha_j$$

$$s.t. \sum_{i=1}^{l} y_i\alpha_i = 0$$

$$0 \le \alpha_i \le C,\ i = 1, \ldots, l.$$

决策函数: $f(x) = \sum_{i=1}^{l} \alpha_i y_i K(x_i, x) + b$

高斯核:

$$K(x_i, x_j) = \exp\left(-\frac{||x_i - x_j||^2}{2\boldsymbol{\sigma}^2}\right)$$

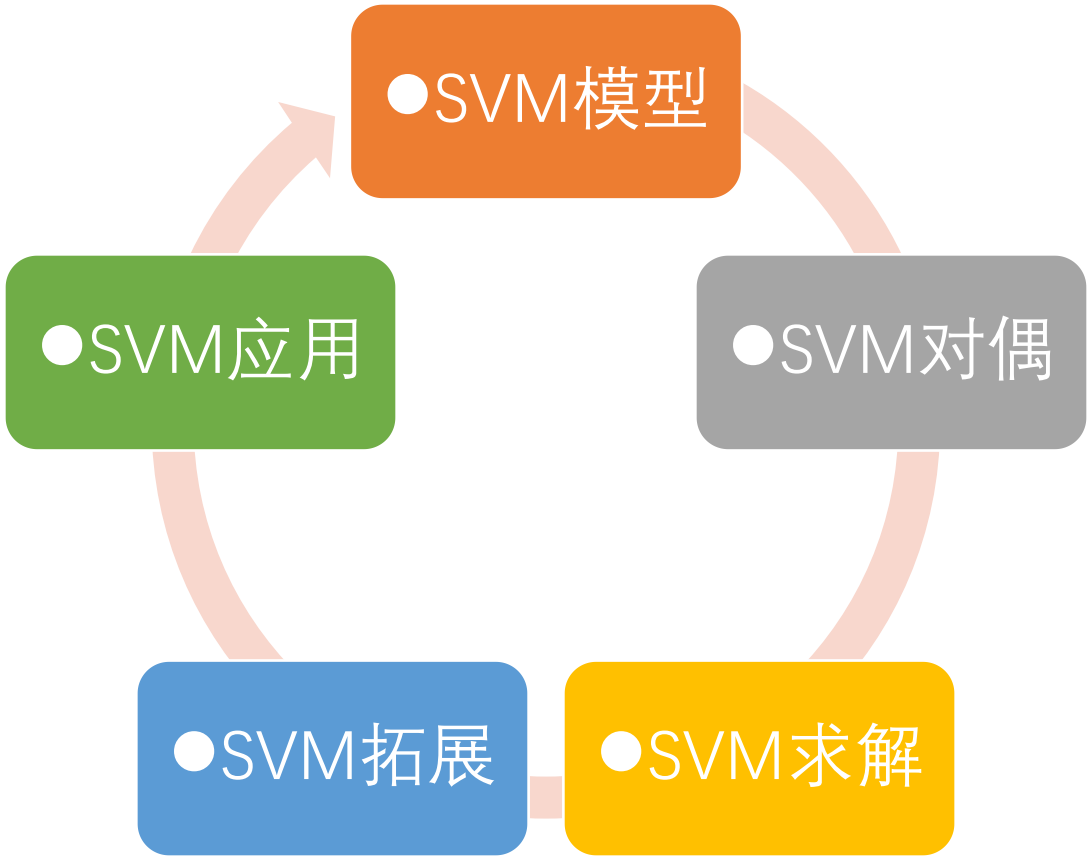
选参

$$\min_{w,b,\xi} \quad \frac{1}{2}\|w\|^2 + \boldsymbol{C}\sum_{i=1}^{l}\xi_i$$

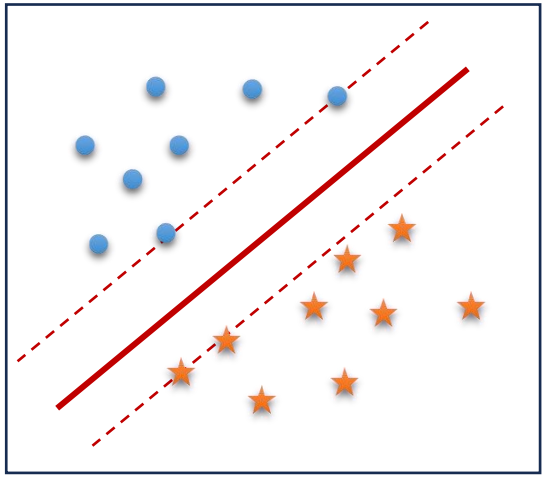
$$s.t.\ y_i(w^T\Phi(x_i) + b) \ge 1 - \xi_i,\ i = 1, \ldots, l$$

$$\xi_i \ge 0,\ i = 1, \ldots, l$$

大纲

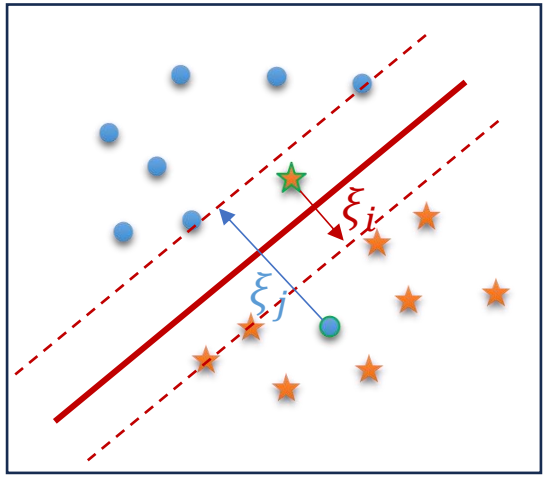


线性可分



$$\min_{w,b} \ \frac{1}{2}||w||^2$$
$$s.t. \quad y_i\big((w \cdot x_i) + b\big) \geq 1, \quad i = 1, \cdots, l$$

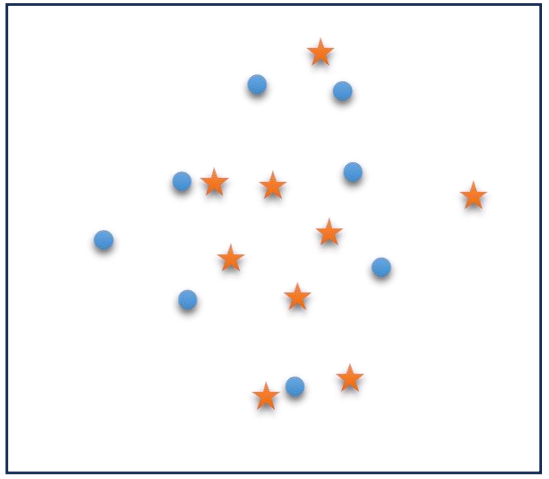
近似线性可分（线性不可分）



$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\, y_i(w^T x_i + b) \geq 1 - \xi_i, \, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j (x_i \cdot x_j)\alpha_i \alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t. \ \sum_{i=1}^{l} y_i \alpha_i = 0$$
$$0 \leq \alpha_i \leq C, \ i = 1, \ldots, l.$$

非线性可分



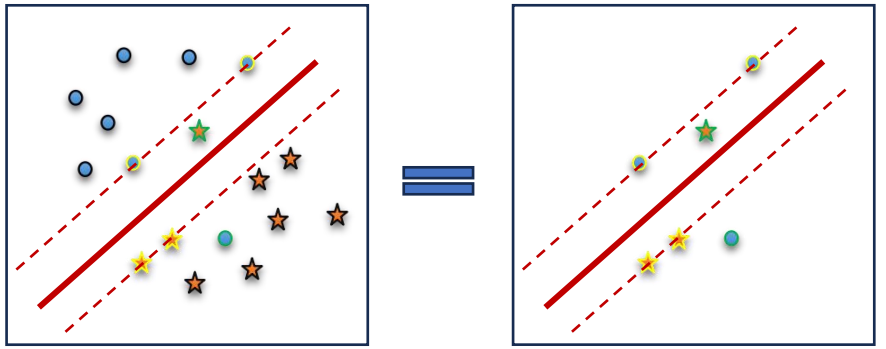
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\, y_i(w^T \Phi(x_i) + b) \geq 1 - \xi_i, \, i = 1, \ldots, l$$
$$\xi_i \geq 0, \ i = 1, \ldots, l$$

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j K(x_i, x_j)\alpha_i \alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t. \ \sum_{i=1}^{l} y_i \alpha_i = 0$$
$$0 \leq \alpha_i \leq C, \ i = 1, \ldots, l.$$

支持向量

Support Vector Machine

支持向量机



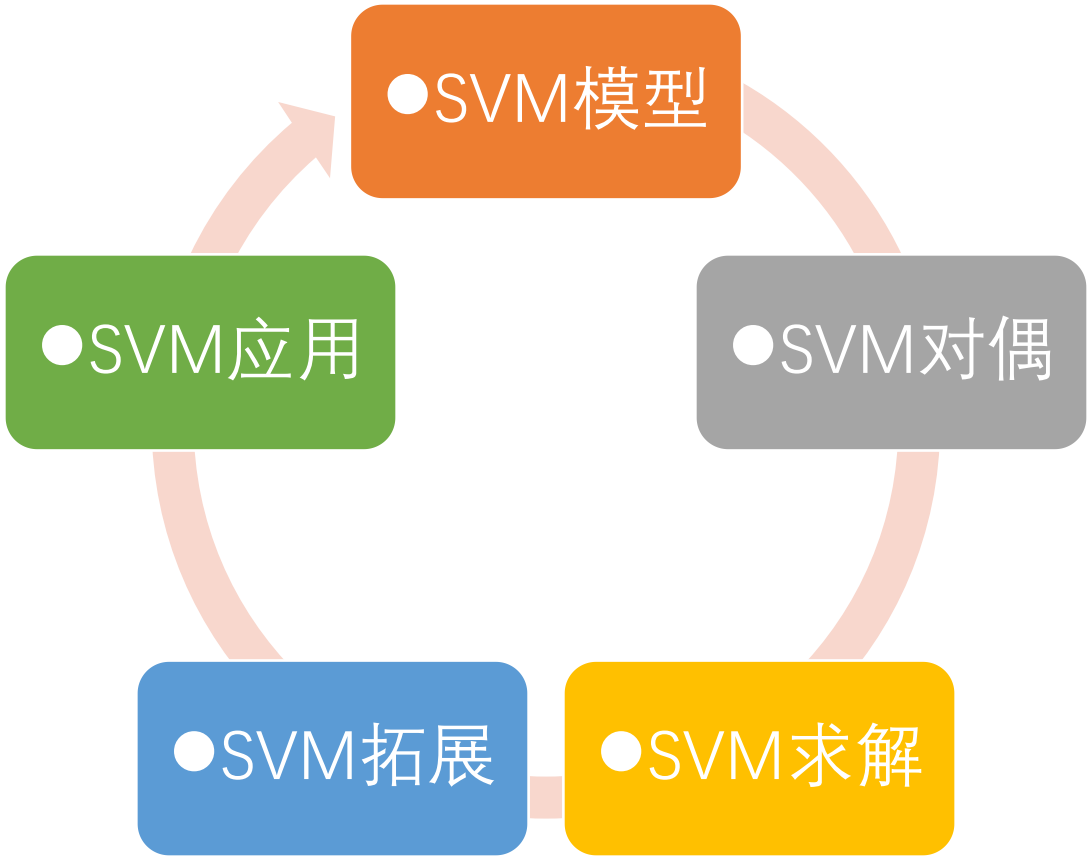
最大间隔

核函数

对偶理论



大纲



SVM求解

原问题

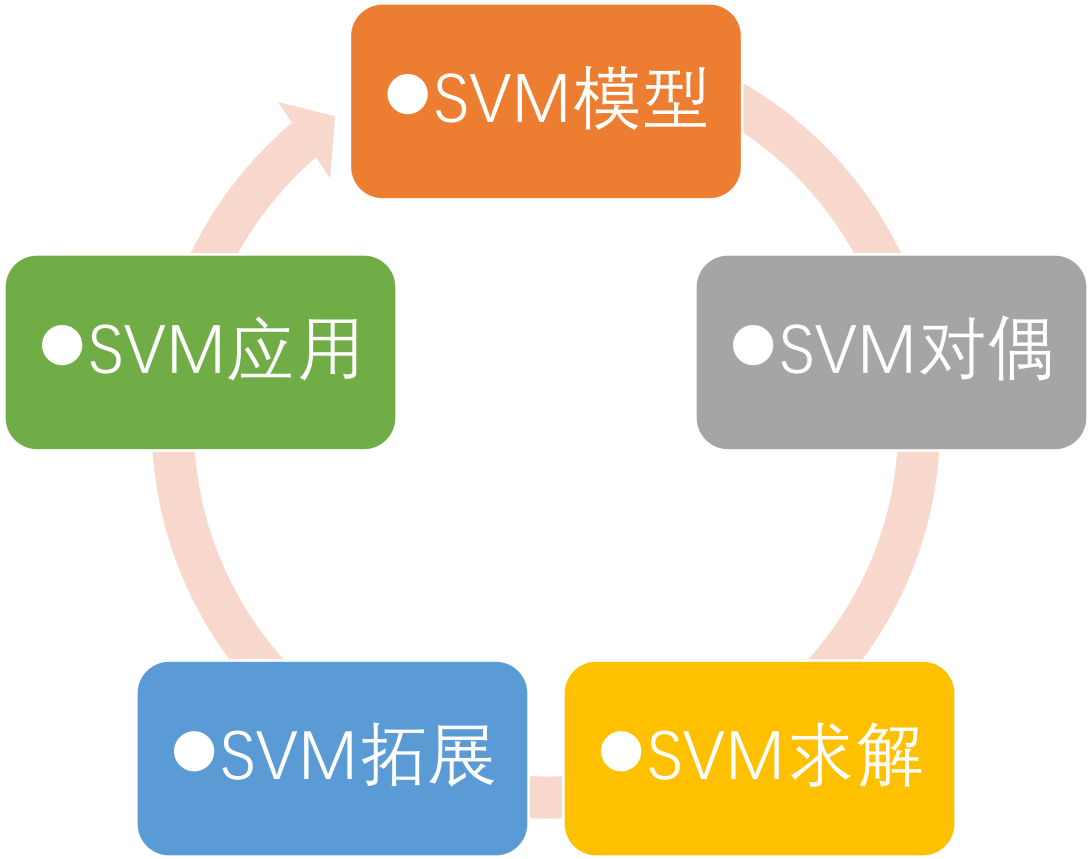
$$\min_{w,b,\xi} \ \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{l}\xi_i$$
$$s.t.\ y_i(w^T\Phi(x_i) + b) \geq 1 - \xi_i, i = 1, \ldots, l$$
$$\xi_i \geq 0,\ i = 1, \ldots, l$$

对偶问题

$$\max_{\alpha} \ -\frac{1}{2}\sum_{i=1}^{l}\sum_{j=1}^{l} y_i y_j K(x_i, x_j)\alpha_i\alpha_j + \sum_{j=1}^{l}\alpha_j$$
$$s.t.\ \sum_{i=1}^{l} y_i\alpha_i = 0$$
$$0 \leq \alpha_i \leq C,\ i = 1, \ldots, l.$$

Algorithm	Citation	SVM type	Optimization type	Style	Runtime
SMO	[Platt, 1999]	Kernel	Dual QP	Batch	$\Omega(n^2 d)$
SVM^{light}	[Joachims, 1999]	Kernel	Dual QP	Batch	$\Omega(n^2 d)$
Core Vector Machine	[Tsang et al., 2005, 2007]	SL Kernel	Dual geometry	Batch	$O(s/\rho^4)$
SVM^{perf}	[Joachims, 2006]	Linear	Dual QP	Batch	$O(ns/\lambda\rho^2)$
NORMA	[Kivinen et al., 2004]	Kernel	Primal SGD	Online(-style)	$\tilde{O}(s/\rho^2)$
SVM-SGD	[Bottou, 2007]	Linear	Primal SGD	Online-style	Unknown
Pegasos	[Shalev-Shwartz et al., 2007]	Kernel	Primal SGD/SGP	Online-style	$\tilde{O}(s/\lambda\rho)$
LibLinear	[Hsieh et al., 2008]	Linear	Dual coordinate descent	Batch	$O(nd \cdot \log(1/\rho))$
SGD-QN	[Bordes and Bottou, 2008]	Linear	Primal 2SGD	Online-style	Unknown
FOLOS	[Duchi and Singer, 2008]	Linear	Primal SGP	Online-style	$\tilde{O}(s/\lambda\rho)$
BMRM	[Smola et al., 2007]	Linear	Dual QP	Batch	$O(d/\lambda\rho)$
OCAS	[Franc and Sonnenburg, 2008]	Linear	Primal QP	Batch	$O(nd)$

大纲




谢 谢
首都师范大学
CAPITAL NORMAL UNIVERSITY